NurturePath AI: Project Overview

Problem, solution, beneficiaries, differentiation, design identity and impact strategy

Prepared as a practical implementation and review document. AI-generated clinical content must not be used as medical advice without qualified review.

01. Executive summary

NurturePath AI is an offline-first maternal healthcare follow-up and referral coordination platform designed for underserved communities. It helps community health workers and clinics maintain continuity of antenatal and postnatal care through appointment reminders, task management, closed-loop referrals, approved educational messaging and privacy-preserving impact analytics. It is designed for Ghana initially and can be adapted to other settings through configurable language, clinic and consent workflows.

Vision: no woman should lose continuity of maternal care simply because connectivity, distance or overstretched systems interrupt follow-up. Mission: equip trusted health workers with accessible tools that turn appointments and referrals into documented completed actions. Tagline: Every mother deserves care that follows her home.

02. Problem statement

A maternal health appointment can be scheduled without a reliable follow-up system that confirms whether a woman was contacted, attended the visit or reached the referred facility. Paper logs, fragmented communication, limited network access and staff workload can make continuity of care difficult. Women may face travel barriers, low digital access and language barriers. The proposed product targets these workflow gaps; it does not claim to solve structural shortages of clinicians, transportation or emergency care.

Evidence requirement: validate the magnitude of each local problem through interviews with adult mothers, midwives and CHWs, and cite reliable public maternal health statistics in the funding application. Do not invent Ghana-specific rates or promise reduced mortality without study evidence.

03. Solution and user value

For health workers: one dashboard of due/overdue appointments, assigned mothers, follow-up outcomes and referrals. For clinics: visibility into referral completion, workload and aggregated service gaps. For women: consent-based reminders and clearer access to follow-up support without requiring an expensive smartphone. For programme funders: defensible metrics that show whether workflows were used and referrals were closed, without publishing identifiable health information.

Unique product hypothesis: 'closed-loop continuity' rather than a generic maternal chatbot. The system tracks each step from scheduling to contact, referral, attendance and closure; offline-first operation makes it useful in low-connectivity environments. This positioning must be tested against existing products and local workflows.

04. How the product works

1. A health worker registers a consenting adult mother with only necessary details and records her communication choices. 2. The worker schedules an antenatal or postnatal follow-up. 3. The system highlights due and overdue tasks, and sends approved neutral reminders when consented. 4. A worker records contact outcomes, reschedules or opens a referral. 5. A clinic team records referral acceptance and completion. 6. The programme dashboard shows aggregated outcomes and bottlenecks.

AI is narrowly used to help prepare simpler-language versions of approved educational materials, with clinical human review. It does not diagnose, triage, prescribe, infer pregnancy risk or replace care professionals.

05. Users and customer model

Primary end users: CHWs, midwives, clinic administrators and consenting pregnant/postpartum adults. Institutional customers may include clinics, healthcare NGOs, maternal health programmes and district-level implementers, subject to procurement and regulatory review. Suggested revenue model: institution-paid subscription or programme licensing and implementation support; essential patient messaging should remain free to beneficiaries wherever funded. Open-source software can coexist with paid hosting, integration, training and support.

06. Product modules

MVP: secure role login; mother enrolment and consent; visit scheduler; follow-up task list; referral status board; offline sync; approved-content message assistant; SMS/WhatsApp simulator; aggregated analytics; audit log; responsive public landing page. Future: validated local-language content, live provider integrations, facility interoperability using appropriate standards, advanced reporting, supervised deployment across regions, patient feedback and partner API integrations. Exclude diagnostic algorithms and unreviewed medical advice.

07. Brand and visual identity

Brand name: NurturePath AI. Symbol: simple heart outline crossed by a small care-path/ECG line, suggesting compassionate continuity, not an official clinical certification. Wordmark: NurturePath in confident rounded sans with AI as subtle modifier. Palette: Deep Navy #123047 (trust, headings), Teal #087E83 (primary actions), Pale Mint #E9F7F4 (supporting backgrounds), Warm Coral #F4A58A (human warmth, non-critical accent), Slate #334A57 (body text), White #FFFFFF (clean surfaces), Red #B42318 (warnings). Typeface: Inter or accessible equivalent. Avoid pink-only stereotyped feminine design; use inclusive real-world imagery only with informed permission.

Logo usage: maintain clear space at least the height of the heart mark; never stretch, rotate or overlay busy photography. Produce full horizontal, icon-only, monochrome and favicon SVGs. No UNICEF branding or suggestion of UNICEF endorsement before any formal award.

08. Responsive experience and accessibility

Design mobile-first for CHWs carrying a phone: 360-430px with thumb-friendly actions, short forms, sticky save state, offline indicator and simplified cards. Tablet: 768-1024px split panels for task lists and details. Desktop: 1280-1440px navigation sidebar, tables, advanced filters and analytics. Support WCAG 2.2 AA goals: semantic forms, keyboard navigation, 44px touch targets, visible focus, screen reader labels, contrast, clear error messages, reduced motion and non-color status indicators. Provide localized dates/timezones and support low-bandwidth environments.

09. Privacy, ethics and clinical safety

Maternal health data is sensitive. Obtain granular consent and lawful processing arrangements; encrypt contact information; apply least-privilege clinic-scoped permissions; use pseudonymous analytics and small-number suppression. Never share identifiable records with public AI tools or publish them as open data. Maintain audit trails and documented deletion/retention rules. Clinical content, referrals and translations require qualified human review; no autonomous diagnosis, risk scoring or emergency handling promises. An actual patient pilot requires legal, clinical and organizational approvals.

10. Outcomes and impact measurement

Measure process outcomes: percentage of scheduled visits with recorded follow-up; percentage of due contacts completed within target window; referral closure rate with explicit denominator; median follow-up delay; opt-out rate; usability task completion; offline sync success. Record baseline and follow-up measures, cohort sizes, exclusions and data completeness. Do not claim reduced maternal mortality, better birth outcomes or improved clinical care without appropriate evidence.

11. Competitive differentiation and validation

Position against generic chatbots and appointment reminder apps: the differentiator is clinic-connected, offline-capable referral closure and accountable follow-up, not AI alone. Validate by interviewing at least several maternal-health professionals and adult women, mapping existing paper workflows, testing with synthetic records, and securing a clinic/NGO letter of interest where feasible. Claims of uniqueness, partnership or impact must be evidence-based.

12. Launch and business roadmap

Stage 1: functional synthetic-data MVP and accessibility review. Stage 2: supervised usability validation and clinician content review. Stage 3: legal/data governance arrangements, secure production infrastructure and limited partner pilot. Stage 4: monitored deployment with feedback, staff training, incident management and published de-identified process indicators. Stage 5: measured expansion and integration with relevant health systems. Initial geography: Ghana; international expansion requires local legal and clinical adaptation.

13. UNICEF application positioning

Problem: missed continuity of maternal care in underserved, low-connectivity settings. Frontier technology: privacy-conscious AI-assisted simplification/translation of approved information plus offline-first sync and workflow analytics; explain honestly which parts use AI and which are deterministic software. Social impact: improved follow-up visibility and referral coordination for women and girls, with initial adult focus.
Open-source plan: release code under a suitable OSI-approved license, excluding secrets and patient records. Financial sustainability: institutional service/support model. Evidence: functioning prototype, initial user testing, clinical review plan and transparent limitations. Check the exact current Cohort 2 call before submitting; selection is not guaranteed.

14. Sample 60-second pitch

In underserved communities, scheduling maternal care is only the beginning. When a woman misses a visit or is referred elsewhere, clinics may struggle to know whether she was contacted or reached care. NurturePath AI helps community health workers close that loop. Our offline-first platform tracks appointments, follow-ups and referrals, while approved multilingual reminders support women without requiring a smartphone. Unlike a generic pregnancy chatbot, we focus on measurable continuity: did the worker follow up, and was the referral completed? We are building with privacy, clinical oversight and open-source reuse from day one. We are seeking partners to validate the workflow and help bring reliable maternal follow-up to communities that need it most.